# Swiggy Restaurant Performance & Market Analysis

## Business Problem

### Background

Swiggy is one of India's leading food delivery platforms, connecting customers with thousands of restaurants across multiple cities. Restaurant performance on the platform can be influenced by factors such as customer ratings, pricing, cuisine type, delivery time, and geographic location.

Understanding these factors is essential to improve customer satisfaction, support restaurant growth, and optimize overall platform performance.

### Problem Statement

Swiggy aims to evaluate restaurant performance across major Indian cities by analyzing customer ratings, pricing patterns, cuisine preferences, delivery efficiency, and customer engagement.

The objective is to identify the key factors influencing customer satisfaction, assess operational performance across cities, understand customer behavior and restaurant trends, and generate data-driven recommendations to improve platform performance and customer experience.

Through exploratory data analysis, statistical analysis, and advanced SQL techniques, this project seeks to uncover actionable insights that support strategic business decision-making.

## Business Objectives

1. Analyze the distribution of restaurants across cities and areas.
2. Identify cities with the highest concentration of restaurants.
3. Evaluate customer rating patterns across restaurants and cuisines.
4. Analyze delivery time performance across different locations.
5. Examine pricing trends across cities and food categories.
6. Identify top-performing restaurants based on ratings and customer engagement.
7. Explore relationships between pricing, ratings, and delivery time.
8. Generate actionable recommendations to improve customer experience and business performance.

## Key Business Questions

- Which cities have the highest number of restaurants?
- Which cities have the highest average restaurant ratings?
- What are the most common cuisine categories on the platform?
- Do higher-priced restaurants receive better ratings?
- How does delivery time vary across cities and areas?
- Which restaurants consistently achieve high ratings?
- What factors appear to influence customer satisfaction?

## Expected Outcome

The analysis will deliver insights into restaurant performance, customer preferences, pricing behavior, and delivery efficiency. These findings can support operational improvements, restaurant onboarding strategies, and customer experience initiatives.

# Dataset Understanding

## Dataset Name: Swiggy Restaurant Dataset

## Source: (Kaggle) https://www.kaggle.com/datasets/abhijitdahatonde/swiggy-restuarant-dataset/data

## Total Records:  Rows 8680, Columns : 10

## Description:

The dataset contains information about restaurants listed on Swiggy including restaurant names, locations, pricing, ratings, cuisine types, and delivery times.

## Columns:

| **Column Name** | **Data Type** |
|---|---|
| ID | int |
| Area | nvarchar |
| City | nvarchar |
| Restaurant | nvarchar |
| Price | float |
| Avg_ratings | float |
| Total_ratings | smallint |
| Food_type | nvarchar |
| Address | nvarchar |
| Delivery_time | tinyint |

# Data Quality Assessment

## Total Rows = 8680

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## Data Types check:

**SQL:**

```
SELECT COLUMN_NAME,
       DATA_TYPE
FROM INFORMATION_SCHEMA.COLUMNS;
```

| Column Name | Data Type |
|---|---|
| ID | int |
| Area | nvarchar |
| City | nvarchar |
| Restaurant | nvarchar |
| Price | float |
| Avg_ratings | float |
| Total_ratings | smallint |
| Food_type | nvarchar |
| Address | nvarchar |
| Delivery_time | tinyint |

**Finding:** Correct

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## Null Value Analysis :

**SQL CODE:**

```
SELECT COUNT(*) - COUNT(ID) AS ID_Nulls,
       COUNT(*) - COUNT(Area) AS Area_Nulls,
       COUNT(*) - COUNT(City) AS City_Nulls,
       COUNT(*) - COUNT(Restaurant) AS Restaurant_Nulls,
       COUNT(*) - COUNT(Price) AS Price_Nulls,
       COUNT(*) - COUNT(Avg_ratings) AS Avg_ratings_Nulls,
       COUNT(*) - COUNT(Total_ratings) AS Total_ratings_Nulls,
       COUNT(*) - COUNT(Food_type) AS Food_type_Nulls,
       COUNT(*) - COUNT(Address) AS Address_Nulls,
       COUNT(*) - COUNT(Delivery_time) AS Delivery_time_Nulls
FROM swiggy;
```

| Column Name | Value |
|---|---|
| ID | 0 |
| Area | 0 |
| City | 0 |
| Restaurant | 0 |
| Price | 0 |
| Avg_ratings | 0 |
| Total_ratings | 0 |
| Food_type | 0 |
| Address | 0 |
| Delivery_time | 0 |

**Decision:** No action required.

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# Blank Values :

We Check Blank Values of these columns **Area, City, Restaurant, Food_type, Address**

```
SELECT
    SUM(CASE WHEN TRIM(Area) = '' THEN 1 ELSE 0 END) AS Area_Blanks,
    SUM(CASE WHEN TRIM(City) = '' THEN 1 ELSE 0 END) AS City_Blanks,
    SUM(CASE WHEN TRIM(Restaurant) = '' THEN 1 ELSE 0 END) AS Restaurant_Blanks,
    SUM(CASE WHEN TRIM(Food_type) = '' THEN 1 ELSE 0 END) AS Food_type_Blanks,
    SUM(CASE WHEN TRIM(Address) = '' THEN 1 ELSE 0 END) AS Address_Blanks
FROM swiggy;
```

**Finding:** Every column return 0

**Decision:** No action required.

Further analysis found that the Price column contained 0 values. Since a price cannot be 0, we investigated further.

**SQL:**

```
SELECT *
FROM swiggy
WHERE Price = 0;
```

**Finding:** 5 restaurant records contained a price value of 0

**Affected Records:**

| ID |
|---|
| 55365 |
| 55710 |
| 73662 |
| 77391 |
| 283144 |

```
SELECT
cast(round((count(*) * 100.0) /(SELECT count(*) FROM swiggy), 2) as decimal(10,2)) AS
Percentage_Zero_Price
FROM swiggy
WHERE Price = 0;
```

Thats **0.06%** of the dataset. Investigation suggests these records likely originated from missing values in the source CSV that were converted to 0 during data import.

**Decision:**

Since a restaurant cannot realistically have a price of 0 and this represented only **0.06%** of the dataset and was not a valid business value. Price values of 0 were treated as missing values and excluded from price-based analysis.

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# Duplicate Check :

**ID Check:**

```
SELECT
    ID,
    COUNT(*)
FROM swiggy
GROUP BY ID
HAVING COUNT(*) > 1;
```

No duplicate Restaurant IDs found.

**Decision:** No action required.

**Exact Duplicate Records Check:**

```
SELECT
    Area,
    City,
    Restaurant,
    Price,
    Avg_ratings,
    Total_ratings,
    Food_type,
    Address,
    Delivery_time,
    COUNT(*)
FROM swiggy
GROUP BY
    Area,
    City,
    Restaurant,
    Price,
    Avg_ratings,
    Total_ratings,
    Food_type,
    Address,
    Delivery_time
HAVING COUNT(*) > 1;
```

**Finding:**

After checking, we found two rows with exactly the same values across all columns.
**City = 'Ahmedabad'**, **Area = 'Kuber Nagar'**, R**estaurant = 'Cake Adda'**.
Therefore, we decided to conduct further analysis before making a decision.

**Further analysis:**

```
SELECT *
FROM swiggy
WHERE ID IN(
    SELECT ID
    FROM swiggy
    WHERE Restaurant = 'Cake Adda'
      AND City = 'Ahmedabad'
      AND Area = 'Kuber Nagar'
);
```

**Finding:**

Identified 2 records (ID= 342814 and 343566) with identical business attributes (Cake Adda, Ahmedabad) but different IDs.

**Decision:** Since the dataset did not provide sufficient information to determine whether these represented true duplicates or separate listings, the records were retained.

# EDA(Exploratory Data Analysis)

## Q1: Which cities have the highest number of restaurants?

### SQL-

```
SELECT   City,
         COUNT(*) AS Restaurant_Count
FROM     Swiggy
GROUP BY City
ORDER BY Restaurant_Count DESC;
```

### Finding:

**Kolkata** leads with **1,346** restaurants — **9.7%** more than **Mumbai (1,227)** and **21.7%** more than **Chennai (1,106)**. Together, the **top 3** cities account for **3,679 restaurants** — **42.4%** of the entire platform's **8,680 listings**, showing high geographic concentration.

### Business Interpretation:

Kolkata appears to be the most concentrated restaurant market within the dataset. The large number of restaurants suggests strong customer demand and high platform adoption. However, restaurant competition may also be higher compared to other cities.

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## Q2: Which cities have the highest average restaurant ratings?

### SQL:

```
SELECT   City,
         COUNT(*) AS Restaurant_Count,
         ROUND(AVG(Avg_ratings), 2) AS Avg_Rating
FROM     Swiggy
GROUP BY City
ORDER BY Avg_Rating DESC;
```

### Finding:

**Chennai** recorded the **highest average restaurant rating (3.78)** among major cities while maintaining a large restaurant base of over **1,100 restaurants**. This suggests consistently strong customer satisfaction across a significant number of restaurants.

**Additional Insight:**

Kolkata
Restaurants = 1346 (Highest)
Rating = 3.70

- **Kolkata** has the **highest number of restaurants** in the dataset while maintaining an average rating of **3.70**, indicating a mature and competitive restaurant ecosystem.

Delhi
Rating = 3.53

- **Delhi's** average rating **(3.53)** sits **0.25** points below Chennai's platform-leading **3.78** — a **7%** satisfaction gap. Closing even half this gap **(0.12 points)** would bring Delhi closer to the platform average of **3.66**, potentially improving customer retention across its restaurant base.

**Business Interpretation:**

Chennai and Bangalore demonstrate stronger customer satisfaction levels compared to other cities in the dataset. Chennai's high rating combined with a large restaurant base suggests a healthy restaurant ecosystem capable of consistently meeting customer expectations.

Kolkata contains the highest number of restaurants while maintaining an average rating of 3.70, indicating strong market maturity and competition.

Delhi reported the lowest average rating (3.53), highlighting a potential opportunity for service quality improvements.

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## Q3: What are the most common cuisine categories on the platform?

**SQL:**

```
SELECT   TOP 20 Food_type,
                COUNT(*) AS Restaurant_Count
FROM     Swiggy
```

```
GROUP BY Food_type
ORDER BY Restaurant_Count DESC;
```

**Finding:**

The top 5 cuisines — **Indian (389)**, **Chinese (277)**, **North Indian (246)**, **Fast Food (240)**, **South Indian (228)** — **account for 1,380 restaurants, representing 15.9%** of all platform listings. Indian cuisine alone holds **4.5%** of total platform share, making it the **single dominant** category by a **40.4%** margin over the **#2 cuisine (Chinese)**.

**Business Interpretation:**

Traditional Indian cuisines dominate the platform, indicating strong customer demand for familiar and locally preferred food options. Chinese and Fast Food categories also maintain a significant presence, reflecting growing demand for quick-service and international cuisine options.

The concentration of restaurants within a few major cuisine categories suggests that these segments represent the core food delivery market and should remain strategic priorities for restaurant acquisition and promotional activities.

## Additional Insight:

**top 5**:

Indian
Chinese
North Indian
Fast Food
South Indian

These are all mass-market cuisines.

**Insight:**

> The dominance of Indian, Chinese, and Fast Food categories indicates that customers primarily prefer familiar and frequently ordered cuisines rather than niche or premium food segments.

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## Q4: Do higher-priced restaurants receive better ratings?

**SQL:**

```
SELECT   CASE WHEN Price < 200 THEN 'Low Price' WHEN Price BETWEEN 200 AND 400 THEN 'M
edium Price' ELSE 'High Price' END AS Price_Category,
         COUNT(*) AS Restaurant_Count,
         ROUND(AVG(Avg_ratings), 2) AS Avg_Rating
FROM     Swiggy
WHERE    Price > 0
GROUP BY CASE WHEN Price < 200 THEN 'Low Price' WHEN Price BETWEEN 200 AND 400 THEN 'M
edium Price' ELSE 'High Price' END
ORDER BY Avg_Rating DESC;
```

**Finding:**

- **Higher-priced** restaurants have the **highest** average rating (**3.79**).
- followed by **low-priced** restaurants **2nd highest** average rating (**3.66**)
- Medium-priced restaurants have the **lowest** average rating (**3.61**).

**Business Interpretation:**

Customers appear to rate premium restaurants more favorably than budget or mid-priced restaurants. This may indicate that higher-priced restaurants provide better food quality, service, packaging, or overall customer experience.

**Additional Insight:**

High Price  = 1787 restaurants
Medium Price = 5911 restaurants

Medium-priced restaurants dominate the platform.

**Price tier distribution — the medium dominance story**

**SQL:**

```
SELECT   CASE WHEN Price < 200 THEN 'Low' WHEN Price <= 400 THEN 'Medium' ELSE 'High'
END AS tier,
         COUNT(*) AS cnt,
         ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER(), 1) AS pct,
         ROUND(AVG(Avg_ratings), 2) AS avg_rating
FROM     Swiggy
WHERE    Price > 0
GROUP BY CASE WHEN Price < 200 THEN 'Low' WHEN Price <= 400 THEN 'Medium' ELSE 'High'
END;
```

💡

> **Medium-priced restaurants (₹200–400)** dominate with **5,911** listings — **68.1%** of the platform. **High-priced restaurants (₹400+)** represent just **20.6% (1,787 listings)**, yet achieve the **highest average rating (3.79)**. This creates a clear quality-price signal: the premium segment outperforms the volume segment by **0.18** rating points despite being **3.3×** smaller in count.

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## Q5: How does delivery time vary across cities and areas?

- **Query 1: Average Delivery Time by City**

**SQL:**

```
SELECT   City,
         ROUND(AVG(Delivery_time), 2) AS Avg_Delivery_Time
FROM     Swiggy
GROUP BY City
ORDER BY Avg_Delivery_Time;
```

**Finding:**

**Ahmedabad** recorded the **fastest** average delivery time **(44 minutes)**, while **Kolkata** recorded the **slowest (67 minutes)**.

**Insight:**

> **Kolkata's average delivery time (67 min) is 52.3% longer than Ahmedabad's (44 min)** — the largest city-level delivery gap on the platform. The platform **average** is **53 min**, meaning **Kolkata** runs **26.4% above average** while **Ahmedabad** runs **16.9% below it.**"

- **Query 2: Delivery Time Distribution:**

**SQL:**

```
SELECT MIN(Delivery_time) AS Min_Time,
       MAX(Delivery_time) AS Max_Time,
```

```
        ROUND(AVG(Delivery_time), 2) AS Avg_Time
FROM    Swiggy;
```

### Finding:

The average delivery time across the platform is **53 minutes**. However, delivery times vary significantly from **20 minutes** to **109 minutes**, indicating substantial operational differences across restaurants and locations.

- **Query 3: Areas with Slowest Delivery**

### SQL:

```
SELECT   TOP 10 Area,
                ROUND(AVG(Delivery_time), 2) AS Avg_Delivery_Time
FROM     Swiggy
GROUP BY Area
ORDER BY Avg_Delivery_Time DESC;
```

### Finding:

**Several Kolkata localities** consistently experience longer delivery times, with **Rabindrapally** averaging **97 minutes**. These areas may face operational challenges such as traffic congestion, delivery partner availability, or restaurant concentration issues.

💡 **Strong Business Insight**

> Earlier, we saw that **Kolkata** has the **highest restaurant count (1,346)**. However, after analyzing overall delivery performance and the areas with the longest delivery times, we found that Kolkata has the **slowest average delivery time**.

### Business Interpretation:

Despite having the highest number of restaurants in the dataset, Kolkata also records the slowest average delivery time. This suggests that restaurant density alone does not guarantee faster deliveries and that operational factors such as traffic conditions, order volume, and delivery network efficiency may significantly impact performance.

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## Q6: Which restaurants consistently achieve high ratings?

**Methodology**

To identify consistently high-performing restaurants, only restaurants with **at least 100 customer ratings** were considered. This helps ensure that high ratings are supported by sufficient customer feedback and are not based on a small number of reviews.

**SQL:**

```
WITH RatedRestaurants
AS (SELECT Restaurant,
           Food_type,
           City,
           ROUND(Avg_ratings, 2) AS Avg_ratings,
           Total_ratings
    FROM Swiggy
    WHERE Total_ratings >= 100)
SELECT TOP 20 *
FROM RatedRestaurants
ORDER BY Avg_ratings DESC, Total_ratings DESC;
```

# Business Insights:

**1. High Ratings Can Be Sustained at Scale**

Restaurants such as:

- Corner House Ice Cream (4.7)
- Mithai (4.7)
- New Kalpana Mistanna Bhander (4.7)

maintain ratings of **4.7+** while receiving **1,000 customer ratings**, demonstrating consistent customer satisfaction at high order volumes.

**2. Kolkata Has Multiple Top-Performing Restaurants**

Several restaurants from Kolkata appear among the top-rated restaurants, including:

- Nic Natural Ice Creams (4.8)

- Mithai (4.7)
- New Kalpana Mistanna Bhander (4.7)
- Mama Mia! - Italian Ice Creams (4.7)

Although Kolkata recorded longer delivery times in earlier analysis, the city also contains multiple highly rated restaurants, suggesting strong food quality and customer satisfaction.

### 3. Specialty category outperformance:

**SQL**:

```
SELECT   Food_type,
         COUNT(*) AS top_count
FROM(SELECT   TOP 20 Food_type
          FROM     Swiggy
          WHERE    Total_ratings >= 100
          ORDER BY Avg_ratings DESC, Total_ratings DESC) AS t
GROUP BY Food_type
ORDER BY top_count DESC;
```

Many top-performing restaurants specialize in:

- Ice Cream
- Desserts
- Bakery
- Coffee

> 💡 Among the **top 20 highest-rated restaurants (min 100 reviews)**, specialty categories — **ice cream, dessert, bakery, coffee** — represent at least **12 of 20 entries (60%+)**, despite these categories being a small fraction of total platform listings. This suggests a **3–4×** higher probability of achieving top-tier ratings for specialty food brands vs mass-market cuisine restaurants.

### Business Interpretation:

Restaurants such as **Nic Natural Ice Creams, Corner House Ice Cream, Mithai, New Kalpana Mistanna Bhander, and Theobroma** emerge as the strongest performers on the platform. Their high ratings combined with substantial customer review volumes indicate consistent service quality and strong customer satisfaction.

The analysis also indicates that customer satisfaction is strongly associated with specialty food categories such as **desserts, ice creams, bakery products, and beverages**. Restaurants within these segments consistently achieve higher ratings and maintain strong customer engagement.

Furthermore, several restaurants have successfully sustained ratings above 4.7 despite receiving hundreds or thousands of customer reviews, making them valuable benchmarks for service quality, product consistency, and customer experience.

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## Q7: What Factors Appear to Influence Customer Satisfaction?

Let's summarize what we've learned so far about customer satisfaction

### Finding 1

```
Chennai = Highest Average Rating (3.78)
```

### Finding 2

```
High Price Restaurants = Highest Ratings (3.79)
```

### Finding 3

```
Top performers are often Ice Cream, Dessert, Bakery brands
```

> So far, we've found that **price appears to influence ratings.**

Now we need to test:

**Delivery Time** vs **Ratings**

**SQL:**

```
SELECT   CASE WHEN Delivery_time < 40 THEN 'Fast' WHEN Delivery_time BETWEEN 40 AND 60
THEN 'Medium' ELSE 'Slow' END AS Delivery_Category,
         COUNT(*) AS Restaurant_Count,
         ROUND(AVG(Avg_ratings), 2) AS Avg_Rating
FROM     Swiggy
GROUP BY CASE WHEN Delivery_time < 40 THEN 'Fast' WHEN Delivery_time BETWEEN 40 AND 60
THEN 'Medium' ELSE 'Slow' END
ORDER BY Avg_Rating DESC;
```

**Major Finding:**

```
Fast Delivery  = 3.82

Medium Delivery = 3.64

Slow Delivery = 3.58
```

**Business Interpretation:**

> **Fast-delivery restaurants (under 40 min) score 3.82** vs **3.58 for slow ones (60+ min)** — a 0.24-point rating gap representing a 6.7% satisfaction advantage. **Only 18.82% of platform restaurants achieve fast delivery**, **meaning 81.2% of listings operate at a rating disadvantage due to delivery speed alone**. If Kolkata improved its delivery to platform average (53 min), its avg rating could move closer to the medium-delivery band (3.64), recovering approximately 0.06 rating points across 1,346 restaurants.

This suggests that **delivery efficiency plays an important role in customer satisfaction**.

**Now Let's Combine Everything:**

**Factor 1: Delivery Speed**

**Evidence:**

```
Fast = 3.82
```

```
Slow = 3.58
```

**Conclusion:**

> Faster delivery contributes positively to customer satisfaction.

---

## Factor 2: Pricing

**Evidence:**

```
High Price = 3.79
Low Price = 3.66
Medium Price = 3.61
```

**Conclusion:**

> Premium restaurants generally achieve higher customer ratings, suggesting customers associate higher prices with better food quality and service.

---

## Factor 3: Restaurant Specialization

**Evidence:**

Top-performing restaurants included:

- Nic Natural Ice Creams (4.8)
- Corner House Ice Cream (4.7)
- Theobroma (4.7)
- Fresh Baked Goodness (4.7)

**Conclusion:**

> Specialty **dessert, bakery, coffee, and ice cream brands** consistently achieve strong customer ratings.

**Final Q7 Answer:**

### Findings

The analysis identified three major factors associated with higher customer ratings:

1. **Faster delivery times**
2. **Higher-priced restaurants**
3. **Specialized food categories such as desserts, bakery products, coffee, and ice cream**

## Business Interpretation

Restaurants delivering food in under 40 minutes achieved the highest average rating (3.82), while restaurants with delivery times exceeding 60 minutes recorded the lowest average rating (3.58). This indicates a strong relationship between delivery efficiency and customer satisfaction.

Additionally, premium-priced restaurants achieved higher average ratings than low-priced and medium-priced restaurants, suggesting that customers may perceive higher-priced restaurants as offering superior food quality and service.

Finally, many of the platform's top-performing restaurants belonged to specialty food categories such as ice cream, bakery, desserts, and coffee, indicating strong customer satisfaction within these segments.

## Conclusion

Customer satisfaction on the **Swiggy** platform appears to be driven by a combination of operational efficiency, perceived quality, and product specialization. Restaurants that deliver quickly, maintain premium offerings, and focus on specialized food categories are more likely to achieve higher customer ratings.

# Advanced Analysis

**1. City Level Performance Analysis**

## Objective

Evaluate the overall performance of each city by analyzing restaurant concentration, pricing, customer satisfaction, customer engagement, and delivery efficiency.

```
WITH CityPerformance AS
(
    SELECT
        City,
        COUNT(*) AS Total_Restaurants,
        ROUND(AVG(Price),2) AS Avg_Price,
        ROUND(AVG(Avg_ratings),2) AS Avg_Rating,
        SUM(Total_ratings) AS Total_Customer_Ratings,
        ROUND(AVG(Delivery_time),2) AS Avg_Delivery_Time
    FROM Swiggy
    WHERE Price > 0
    GROUP BY City
)

SELECT *,
       RANK() OVER
(
           ORDER BY Avg_Rating DESC
       ) AS City_Rank
FROM CityPerformance;
```

### Findings

- Chennai ranked **#1 in customer satisfaction** with an average rating of **3.78**, while maintaining a large restaurant base of **1,105 restaurants**.
- Kolkata had the **largest restaurant presence (1,346 restaurants)** but also recorded the **slowest average delivery time (67 minutes)**.
- Kolkata's delivery time was **52.3% slower than Ahmedabad (44 minutes)**, highlighting potential operational inefficiencies.
- Mumbai emerged as the **most premium market**, with the highest average restaurant price (**₹393.79**), though higher pricing did not directly translate into higher ratings.

- Hyderabad generated the **highest customer engagement**, recording **330,270 customer ratings**, which was **50.3% higher than Kolkata** despite having fewer restaurants.
- Delhi ranked last in customer satisfaction with an average rating of **3.53**, while Chennai outperformed Delhi by approximately **7.1%**.

#### 2. Top-Performing Restaurants Within Each City

For this analysis, we selected restaurants with a minimum of 100 total ratings.

```
WITH RankedRestaurants AS
(
    SELECT
        City,
        Restaurant,
        Avg_ratings,
        Total_ratings,
        ROW_NUMBER() OVER
(
            PARTITION BY City
            ORDER BY Avg_ratings DESC,
                     Total_ratings DESC
        ) AS rn
    FROM Swiggy
    WHERE Total_ratings >= 100
)

SELECT *
FROM RankedRestaurants
WHERE rn <= 5;
```

## Findings

- **Nic Natural Ice Creams** emerged as a top performer in multiple cities, ranking #1 in Kolkata and Surat, demonstrating strong brand consistency across locations.
- **Corner House Ice Cream (Bangalore)** and **Mithai (Kolkata)** stood out with **1,000 customer ratings**, indicating sustained customer satisfaction at scale.
- Dessert, ice cream, bakery, and specialty food brands dominated the top rankings across almost every city.

- Chennai showed remarkable consistency, with all top 5 restaurants maintaining ratings of **4.7** and **500 customer ratings** each.
- Mumbai's highest-ranked restaurant, **Blue Tokai Coffee Roasters**, achieved a rating of **4.8**, highlighting strong customer preference for premium coffee and specialty food brands.
- Several top-performing restaurants belonged to well-established chains such as **Theobroma**, **Natural Ice Cream**, **Nic Natural Ice Creams**, and **Kwality Walls**, suggesting that brand reputation contributes positively to customer satisfaction.

### 3. Statistical Distribution

**Price Distribution:**

```
SELECT
    MIN(Price) AS Min_Price,
    MAX(Price) AS Max_Price,
    ROUND(AVG(Price),2) AS Avg_Price,
    ROUND(STDEV(Price),2) AS Std_Dev_Price
FROM Swiggy
WHERE Price > 0;
```

**Findings:**

- Restaurant prices ranged from **₹1 to ₹2,500**, indicating a wide variety of pricing options on the platform.
- The average restaurant price was **₹348.65**, while the standard deviation was **₹230.86**, showing significant variation in pricing.
- A small number of premium-priced restaurants exist, with the highest-priced restaurants charging over **7 times** the average price.
- Only **15 restaurants (0.17% of the dataset)** had prices below ₹10, suggesting potential anomalies or special pricing cases.

**Rating Distribution:**

```
SELECT
    MIN(Avg_ratings) AS Min_Rating,
```

```
    MAX(Avg_ratings) AS Max_Rating,
    ROUND(AVG(Avg_ratings),2) AS Avg_Rating,
    ROUND(STDEV(Avg_ratings),2) AS Std_Dev_Rating
FROM Swiggy;
```

**Findings:**

- Restaurant ratings ranged from **2.0 to 5.0**, indicating varying levels of customer satisfaction across the platform.
- The average restaurant rating was **3.66**, suggesting that most restaurants maintain moderate to good customer satisfaction.
- The standard deviation of **0.65** is relatively low compared to the rating scale, indicating that ratings are fairly concentrated around the average.
- Most restaurants appear to operate within a similar rating range, with relatively few extreme high- or low-performing restaurants.

**Delivery Time Distribution:**

```
SELECT
    MIN(Delivery_time) AS Min_Delivery,
    MAX(Delivery_time) AS Max_Delivery,
    ROUND(AVG(Delivery_time),2) AS Avg_Delivery,
    ROUND(STDEV(Delivery_time),2) AS Std_Dev_Delivery
FROM Swiggy;
```

**Findings:**

- Delivery times ranged from **20 to 109 minutes**, showing considerable variation in service speed across restaurants.
- The average delivery time was **53 minutes**.
- The standard deviation of **14.29 minutes** indicates moderate variability in delivery performance.
- The slowest deliveries took more than **5 times longer** than the fastest deliveries, highlighting operational differences across locations and restaurants.

## Median:

### Median Price:

```
SELECT DISTINCT
    PERCENTILE_CONT(0.5)
    WITHIN GROUP(ORDER BY Price)
    OVER() AS Median_Price
FROM Swiggy
WHERE Price > 0;
```

### Median Rating:

```
SELECT DISTINCT
    PERCENTILE_CONT(0.5)
    WITHIN GROUP(ORDER BY Avg_ratings)
    OVER() AS Median_Rating
FROM Swiggy;
```

### Median Delivery Time:

```
SELECT DISTINCT
    PERCENTILE_CONT(0.5)
    WITHIN GROUP(ORDER BY Delivery_time)
    OVER() AS Median_Delivery_Time
FROM Swiggy;
```

## Price Distribution Analysis:

### Findings:

- Restaurant prices vary significantly, ranging from **₹1 to ₹2500**.
- The average price (**₹348.65**) is **16.2% higher** than the median price (**₹300**), indicating a right-skewed distribution.
- The standard deviation (**₹230.86**) represents **66.2% of the average price**, highlighting substantial pricing variability across restaurants.

- Premium restaurants significantly influence the overall average price.

## Customer Rating Distribution Analysis:

**Findings:**

- Customer ratings range from **2.0 to 5.0**, covering the full rating spectrum.
- The median rating (**3.90**) is **6.7% higher** than the average rating (**3.66**), suggesting that a relatively small number of poorly rated restaurants lower the platform average.
- The standard deviation (**0.65**) represents only **17.8% of the average rating**, indicating low variability in customer satisfaction.
- Most restaurants are clustered around ratings between 3.5 and 4.5.

## Delivery Time Distribution Analysis:

**Findings:**

- Delivery times range from **20 to 109 minutes**, showing considerable operational variation.
- The average and median delivery times are identical (**53 minutes**), indicating a relatively balanced distribution.
- The standard deviation (**14.29 minutes**) accounts for **27.0% of the average delivery time**, suggesting moderate delivery variability.
- The slowest deliveries take **445% longer** than the fastest deliveries.

# KPI Definition

| KPI | Definition / Formula | Business Purpose |
| --- | --- | --- |
| Total Restaurants | Count of all restaurants in the dataset | Measures the overall size of the restaurant marketplace |
| Average Rating | Average of Avg_ratings | Measures overall customer satisfaction |
| Average Delivery Time | Average of Delivery_time | Measures operational efficiency and delivery performance |
| Fast Delivery Percentage | Percentage of restaurants with delivery time less than 40 minutes | Measures delivery speed performance |
| High Rated Restaurants | Count of restaurants with rating greater than or equal to 4.5 | Measures the number of top-performing restaurants |
| Total Cities | Count of distinct cities | Measures platform geographic coverage |
| Average Price | Average restaurant price excluding invalid values | Measures overall pricing level of restaurants on the platform |

## 1. Customer Satisfaction KPIs

- **Average Rating**

SQL:

```
SELECT ROUND(AVG(Avg_ratings), 2) AS Avg_Rating
FROM   Swiggy;
```

**Avg Rating : 3.66**

💡 The platform average rating is **3.66** out of **5 (73.2% score)**. With the highest-rated city **(Chennai)** at **3.78** and **lowest (Delhi)** at **3.53**, there is a **0.25**-point spread across cities — suggesting inconsistent customer experience depending on location.

- **High Rated Restaurants**

**SQL:**

```
SELECT COUNT(*) AS High_Rated_Restaurants
FROM   Swiggy
WHERE  Avg_ratings >= 4.5;
```

**High Rated Restaurants : 662**

💡 Only **662** of **8,680** restaurants **(7.6%)** achieve a rating of **4.5** or above — meaning **92.4%** of the platform's restaurants fall below the high-performance threshold. This identifies a large opportunity for quality improvement across the majority of listed partners.

## 2. Operational KPIs:

- **Average Delivery Time**

**SQL:**

```
SELECT ROUND(AVG(Delivery_time), 2) AS Avg_Delivery_Time
FROM   Swiggy;
```

**Average Delivery Time: 53 min**

- **Fast Delivery Percentage**

**SQL:**

```
SELECT Concat(CAST(COUNT(CASE WHEN Delivery_time < 40 THEN 1 END) * 100.0 / COUNT(*) A
S DECIMAL(10, 2)), ' %') AS Fast_Delivery_Percentage
FROM   Swiggy;
```

- **Percentage of restaurants providing fast delivery was : 18.82 %**

## 3. Marketplace KPIs:

- **Total Restaurants**

**SQL:**

```
SELECT COUNT(*) AS Total_Restaurants
FROM   Swiggy;
```

**Total Restaurants: 8680**

- **Total Cities**

**SQL:**

```
SELECT COUNT(DISTINCT City) AS Total_Cities
FROM   Swiggy;
```

**Total Cities: 9**

- **Average Price:**

**SQL:**

```
SELECT ROUND(AVG(Price), 2) AS Avg_Price
FROM   Swiggy
WHERE  Price > 0;
```

**Average Price: 348.65/-**

💡 The platform average price is **₹348.65** — **sitting in the upper half of the medium-price band (₹200–400),** just **13.4% below the ₹400 premium threshold.** This suggests most restaurants are priced near the premium boundary, with a natural upgrade opportunity for quality improvements.

# Business Insights Summary

## Insight 1: Kolkata Dominates Restaurant Availability but Faces Delivery Challenges

Kolkata recorded the highest number of restaurants (1,346), which is 21.8% higher than Chennai (1,105). However, Kolkata also recorded the slowest average delivery time (67 minutes), indicating operational challenges despite its strong restaurant presence.

---

## Insight 2: Chennai Leads Customer Satisfaction

Chennai achieved the highest average restaurant rating (3.78), ranking #1 among all cities. Compared to Delhi (3.53), Chennai's average rating was approximately 7.1% higher, suggesting stronger overall customer satisfaction and service quality.

---

## Insight 3: Delivery Speed Has a Measurable Impact on Ratings

Restaurants delivering within 40 minutes achieved an average rating of 3.82, compared to 3.58 for restaurants with delivery times exceeding 60 minutes.

This represents a 6.7% improvement in customer ratings, demonstrating that delivery efficiency is a key driver of customer satisfaction.

---

## Insight 4: Premium Restaurants Receive Higher Customer Ratings

High-priced restaurants achieved the highest average rating (3.79), outperforming medium-priced restaurants (3.61) by approximately 5.0%.

This suggests that customers associate premium pricing with higher food quality, better service, and an improved overall dining experience.

---

## Insight 5: Hyderabad Demonstrates Exceptional Customer Engagement

Hyderabad generated 330,270 customer ratings, the highest among all cities in the dataset.

Despite having fewer restaurants than Kolkata, Hyderabad generated 50.3% more customer ratings, indicating stronger customer engagement and ordering activity.

---

## Insight 6: Specialty Food Brands Consistently Outperform

Many of the platform's top-performing restaurants belonged to ice cream, dessert, bakery, and coffee categories.

Brands such as Nic Natural Ice Creams, Natural Ice Cream, Theobroma, and Krispy Kreme consistently achieved ratings above 4.5, indicating strong customer satisfaction within specialized food segments.

---

## Insight 7: Nic Natural Ice Creams Emerged as the Strongest Multi-City Brand

Nic Natural Ice Creams maintained an average rating of 4.59 across 9 cities while accumulating over 4,000 customer ratings.

This demonstrates exceptional brand consistency, operational standardization, and customer satisfaction across multiple markets.

---

## Insight 8: Restaurant Pricing Shows Significant Variability

Restaurant prices ranged from ₹1 to ₹2500, with an average price of ₹348.65 and a median price of ₹300.

The average price was 16.2% higher than the median, indicating a right-skewed pricing distribution where a relatively small number of premium restaurants increase the overall average.

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## Insight 9: Customer Ratings Are More Consistent Than Pricing

While pricing exhibited high variability, customer ratings remained relatively stable.

The average rating was 3.66 with a standard deviation of only 0.65, suggesting that most restaurants deliver a broadly similar customer experience despite significant differences in pricing and delivery performance.

# Recommendations

## Recommendation 1: Improve Delivery Efficiency in Kolkata

### Based On

- Kolkata has the highest number of restaurants.
- Kolkata has the slowest average delivery time (67 minutes).

### Recommendation

> Swiggy should investigate delivery operations in Kolkata and identify bottlenecks affecting delivery speed. Optimizing delivery partner allocation and improving route efficiency may help reduce delivery times and improve customer satisfaction.

---

## Recommendation 2: Promote High-Performing Restaurant Categories

### Based On

- Ice cream, dessert, bakery, and coffee brands consistently achieved high ratings.

### Recommendation

> Swiggy should increase visibility for high-performing specialty food categories through featured listings, promotional campaigns, and personalized recommendations.

---

## Recommendation 3: Encourage Faster Deliveries

### Based On

```
Fast Delivery Rating = 3.82
```

```
Slow Delivery Rating = 3.58
```

### Recommendation

> Swiggy should establish delivery-time performance benchmarks and incentivize restaurants and delivery partners to maintain faster delivery times.

---

## Recommendation 4: Learn from Top-Performing Restaurants

### Based On

- Nic Natural Ice Creams
- Corner House Ice Cream
- Mithai
- Theobroma

consistently achieved high ratings.

### Recommendation

> Swiggy should analyze operational practices of top-performing restaurants and identify best practices that can be shared with lower-performing restaurant partners.

---

## Recommendation 5: Replicate Chennai's Success

### Based On

- Chennai recorded the highest average rating. **(3.78)**

### Recommendation

> Chennai ranked #1 in customer satisfaction with an average rating of **3.78**, outperforming Delhi by approximately **7.1%**. Swiggy should analyze

> operational practices, restaurant mix, and customer experience factors in Chennai and replicate successful strategies in lower-performing cities.

---

# Recommendation 6: Focus on High-Engagement Markets

## Based On

```
Hyderabad = 330,270 customer ratings
Highest customer engagement
50.3% higher than Kolkata
```

## Recommendation

> Hyderabad demonstrates exceptionally strong customer engagement. Swiggy should prioritize restaurant acquisition, loyalty programs, and premium offerings in high-engagement markets to maximize revenue opportunities.

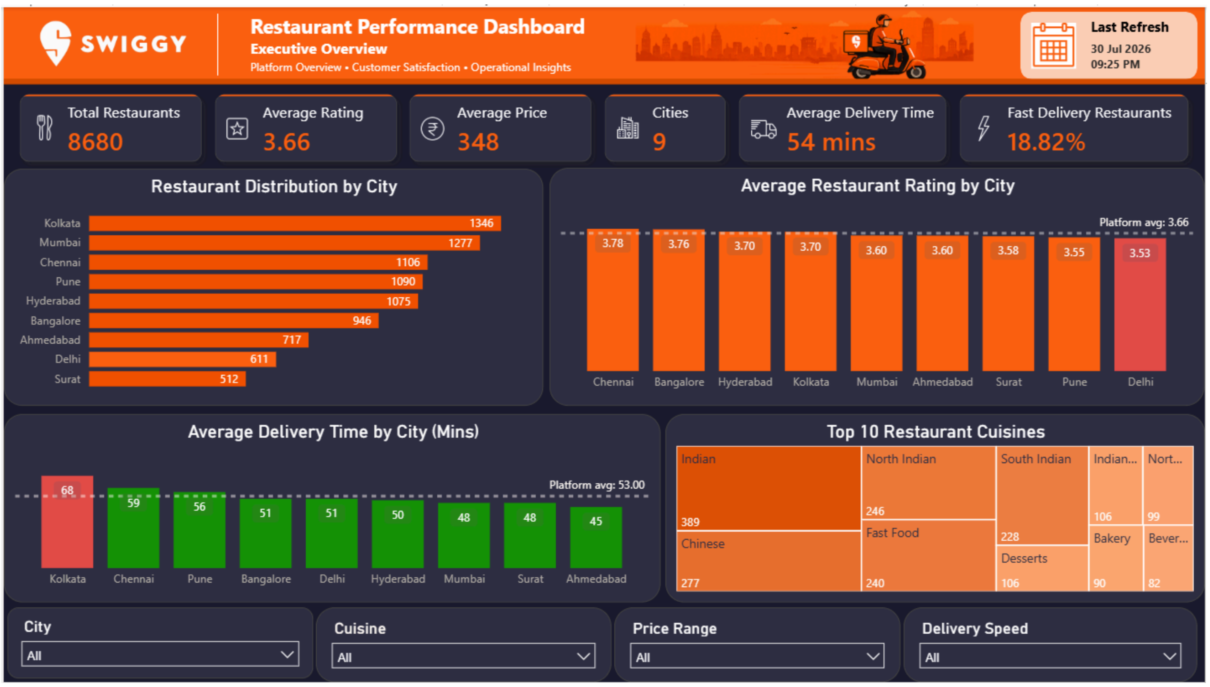
# Power Bi Dashboard

## 📊 Executive Overview Dashboard

Purpose: Provides a high-level overview of the Swiggy restaurant ecosystem.

### Highlights

Total Restaurants • Average Rating • Average Delivery Time • Average Price • Fast Delivery % • Restaurant Distribution by City • Cuisine Analysis • Delivery Performance • Pricing Insights




## 🍽️ Restaurant Performance Dashboard

Purpose: Helps identify the best-performing restaurants and customer engagement patterns.

### Highlights

Highest Rated Restaurant • Most Popular Restaurant • Top Rated City • Restaurants Rated 4.5+ • Top 10 Most Popular Restaurants • Restaurant

Performance Leaderboard • Delivery Speed vs Customer Rating • Rating by Price Category